

Quantum Technologies Research Project

March Report

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Project Goals & Metrics

Research Gap: No unified comparison of BB84 vs MDI-QKD across key metrics

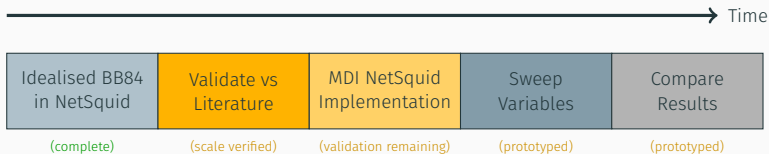
**Key Rate vs
Distance**
(varying networks)

**Implementation
Cost**
(economic & energetic)

Scalability
(user count)

- Analyse both protocols across varying network topologies
- Identify parameter regimes & topologies where each excels
- Inform practical QKD deployment decisions

Recap: Where We Were



Implementation stages:

1. Finalise BB84 implementation in NetSquid
2. Validate against known rate-distance curves from literature
3. Adjust BB84 → MDI-QKD with symmetric setup (+ validate)
4. Sweep distances, topologies and scale
5. Comparative analysis across metrics

Since last meeting

March Progress

- Loss model: Beer-Lambert $T = 10^{-\alpha L/10}$ implemented
- Detector efficiency model implemented
- Three comparative sweep graphs produced
- Literature Review submitted

Action list

- Relative *and* absolute key rate scales + MDI duplicated distance 
- Effect of BB84 repeaters on comparison 
- Vary Charlie's location 
- Practical key rate thresholds 
- One-pager: modelling plan & layering inc. MDI synchronisation 

Modelling Status: Layered Build-up

Layer 1: Fibre Loss; $T = 10^{-\alpha L/10}$	complete
Layer 2: Detector Efficiency; $\eta_d \in [0.5, 1.0]$	complete
Layer 3: Dark Counts	next
Layer 4: Node / Connector Loss	planned
Layer 5: Source Bit Errors	planned
Layer 6: Detector Basis Bias	planned
Layer 7: Fibre de-phasing	planned

Note: Error correction and privacy amplification are *not* planned - both protocols use identical methods, so omitting them preserves fairness of comparison at the physical layer.

Results I: Key Rate vs Fibre Attenuation Rate

- Fixed: 50 km Alice–Bob distance, $\eta_d = 0.9$
- Both curves **linear on log scale** $\Rightarrow$ exponential decay in rate with attenuation
- MDI offset below BB84 by $\approx$ constant factor across all α
- Offset consistent with 50% BSM efficiency cap and η_d^2 dependence
- At $\alpha = 0.2$ dB/km (standard SMF): BB84 ≈ 70 kbps, MDI ≈ 25 kbps
- Both remain operationally viable across commercial attenuation range

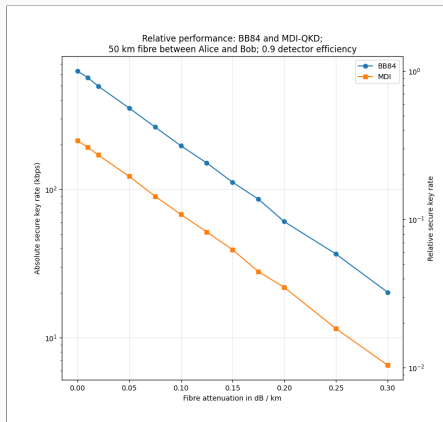


Figure 1: Key rate vs fibre attenuation. 50 km, $\eta_d = 0.9$.

Results II: Key Rate vs Distance

- Fixed: $\alpha = 0.2 \text{ dB/km}$, $\eta_d = 0.9$
- BB84 and MDI both show near-linear decay on log scale at long distances
- MDI decays **faster**: gap widens beyond $\sim 30 \text{ km}$
- At 100 km : BB84 $\sim 4 \text{ kbps}$, MDI $\sim 2 \text{ kbps}$
- Crossover not observed in this range

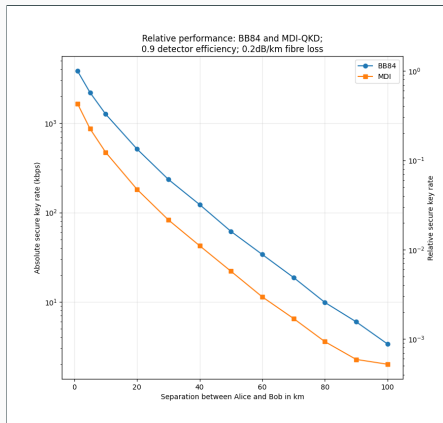


Figure 2: Key rate vs Alice–Bob separation.
 $\alpha = 0.2 \text{ dB/km}$, $\eta_d = 0.9$.

Results III: Key Rate vs Detector Efficiency

- Fixed: 50 km, $\alpha = 0.2$ dB/km
- BB84 degrades **slowly**: near-linear in η_d , losing $\sim 50\%$ from $\eta_d = 1 \rightarrow 0.5$
- MDI degrades **quadratically**: losing $\sim 90\%$ over the same range
- MDI depends on coincident detection at two detectors
- **Practical implication:** MDI is strongly penalised by imperfect detectors
- SNSPDs ($\eta_d \approx 0.9\text{--}0.95$) are therefore *not* optional for MDI

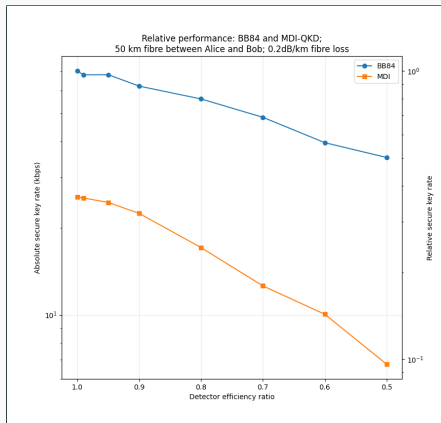
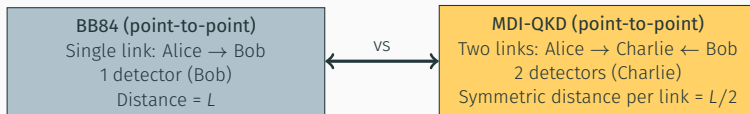


Figure 3: Key rate vs detector efficiency. 50 km, $\alpha = 0.2$ dB/km.

Discussion: Fairness of Comparison

Concurrent question: Are MDI and BB84 being compared like for like?



- Current: BB84 and MDI use identical Alice–Bob separation L , and **all sources and detectors are the same**
- MDI disadvantage in rate arises from: (1) 50% BSM efficiency cap, (2) η_d^2 at Charlie's two detectors
- Greatest effect on key rate decay (for both) is fibre attenuation - **3 orders of magnitude** drop-off from 1 to 100 km.
- Constant ratio exhibited for distance and attenuation - detector efficiency causes a **linear increase in gap** between BB84 and MDI

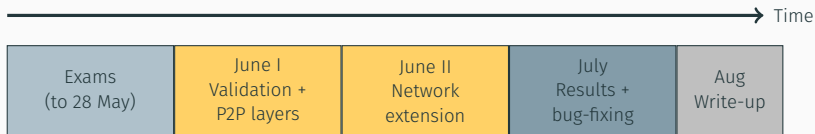
Simulation priorities (June):

- Dark count model: scale d_c proportionally with detection rate; sensitivity analysis on d_c/η ratio
- Build in further modelling parameters
- Begin validation: match absolute key rate curves
- Start work extending to network-scale simulation

Analysis priorities:

- Modelling proposal - device and network parameters
- Identify QBER threshold crossings ($> 11\%$) as a function of distance, loss, and η_d for trusted node analysis
- Assess whether Charlie's position affects performance

Timeline



- **June first half:** Validation benchmarking (BB84 + MDI); complete P2P modelling layers (dark counts → node loss → beyond)
- **June second half:** Begin network-scale extension; trusted node BB84 vs MDI comparison
- **July:** Inevitable bug-catching, parameter sweeps, result collection
- **August:** Write-up, final figures, submission preparation; submission
- **September:** Viva